The Materials Selection Process

1. Pick Application ⟶ Determine required Properties

   Properties: mechanical, electrical, thermal, magnetic, optical, deteriorative.

2. Properties ⟶ Identify candidate Material(s)

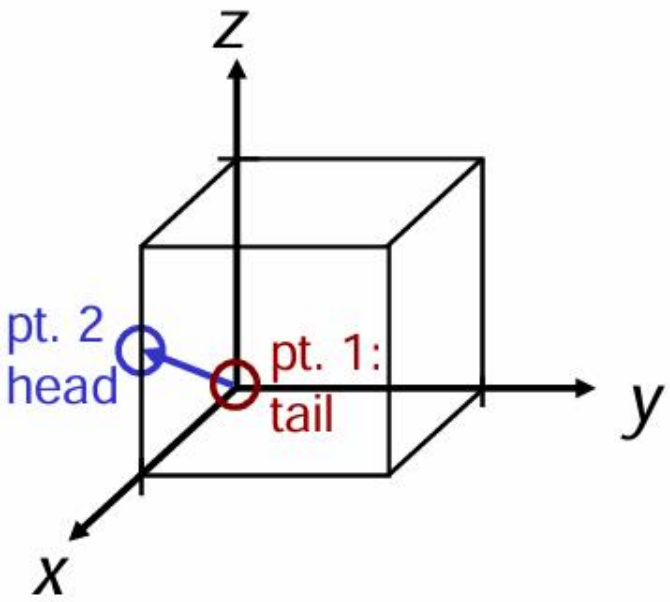
   Material: structure, composition.

3. Material ⟶ Identify required Processing

   Processing: changes *structure* and overall *shape*
   ex: casting, sintering, vapor deposition, doping
   forming, joining, annealing.

WILEY

Crystallographic Directions



Algorithm

1. Determine coordinates of vector tail, pt. 1: x_1, y_1, & z_1; and vector head, pt. 2: x_2, y_2, & z_2.
2. Tail point coordinates subtracted from head point coordinates.
3. Normalize coordinate differences in terms of lattice parameters *a*, *b*, and *c*:

$$\frac{x_2 - x_1}{a} \quad \frac{y_2 - y_1}{b} \quad \frac{z_2 - z_1}{c}$$

4. Adjust to smallest integer values
5. Enclose in square brackets, no commas

[*uvw*]

ex:

pt. 1 $x_1 = 0$, $y_1 = 0$, $z_1 = 0$

pt. 2 $x_2 = a$, $y_2 = 0$, $z_2 = c/2$

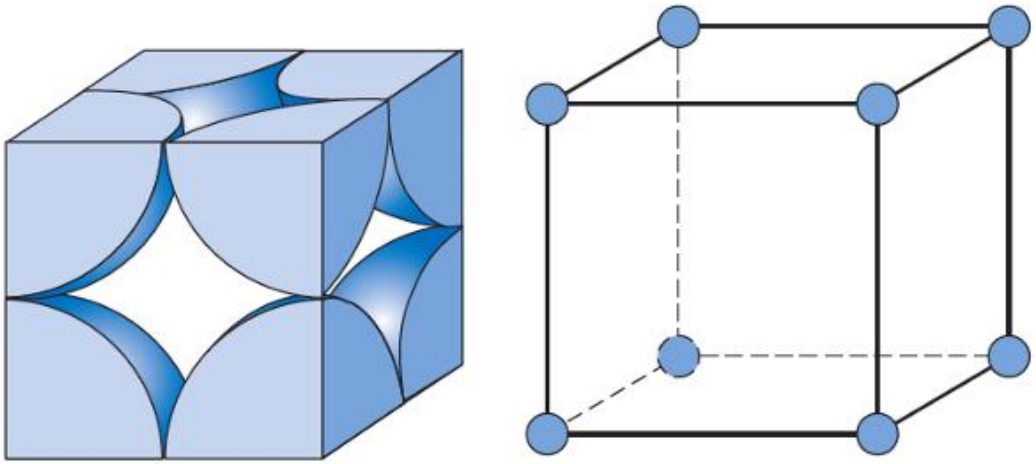
$$\frac{a-0}{a} \quad \frac{0-0}{b} \quad \frac{c/2-0}{c}$$

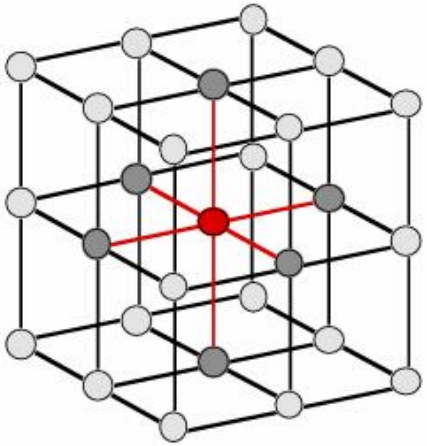
=> 1, 0, 1/2 => 2, 0, 1

=> [201]

Simple Cubic Structure (SC)

- Rare due to low packing density (only Polonium (Po=84) has this structure)
- Close-packed directions are cube edges.
- Coordination # = 6 (# nearest neighbors)





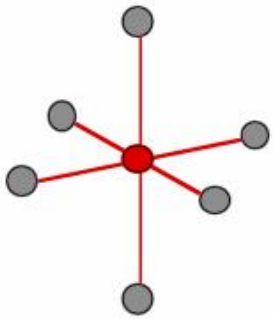


Fig. 4.2, *Callister & Rethwisch 9e.*

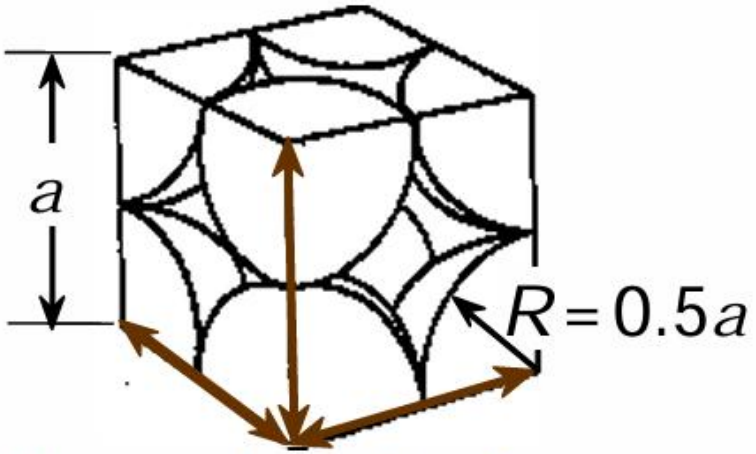
WILEY

Atomic Packing Factor (APF)

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

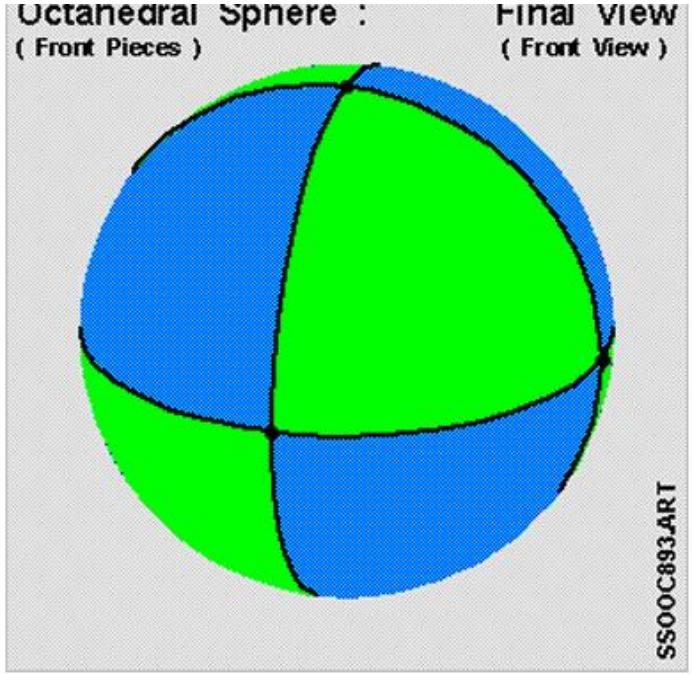
contains 8 x 1/8 =

1 atom/unit cell

$$APF = \frac{\text{atoms}}{\text{unit cell}} \cdot \frac{\text{volume}}{\text{atom}} \Big/ \frac{a^3}{\text{volume}}{\text{unit cell}}$$

$$APF = \frac{1 \cdot \frac{4}{3}\pi (0.5a)^3}{a^3}$$

Adapted from Fig. 4.2 (a), *Callister & Rethwisch 9e.*



$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell =
Number of atoms X Volume of atom

for SC Number of atom = close-packed directions =

contains 8 x 1/8 = 1 atom/unit cell

But a = 2r

Volume of atom = $\frac{4}{3}\pi(0.5a)^3 = \frac{4}{3}\pi r^3$

Volume of Unit cell = $a^3 = (2r)^3 = 8r^3$

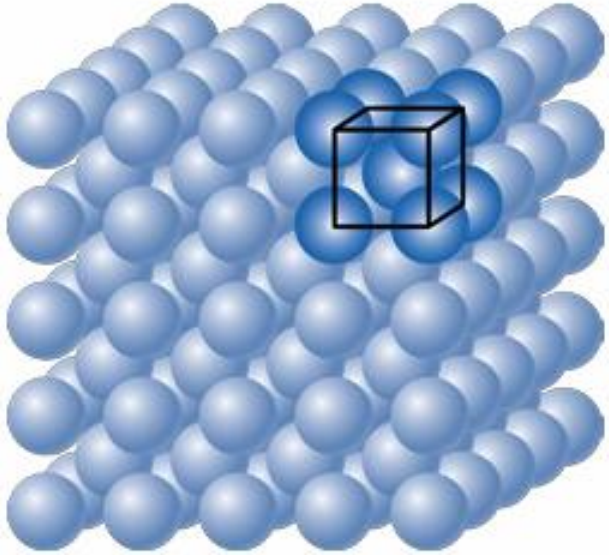
$$APF\ for\ SC = \frac{1 \times \frac{4}{3}\pi r^3}{(2r)^3} = \frac{\frac{4}{3}\pi r^3}{8r^3} = \frac{\frac{4}{3}\pi}{8} = 0.52$$

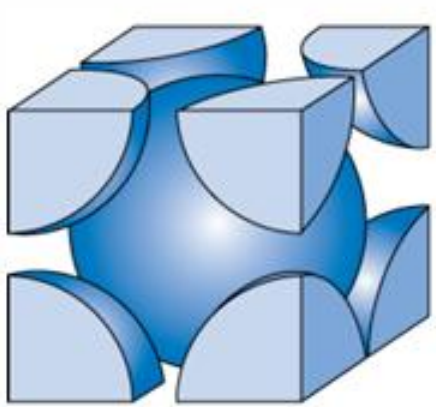
Body Centered Cubic Structure (BCC)

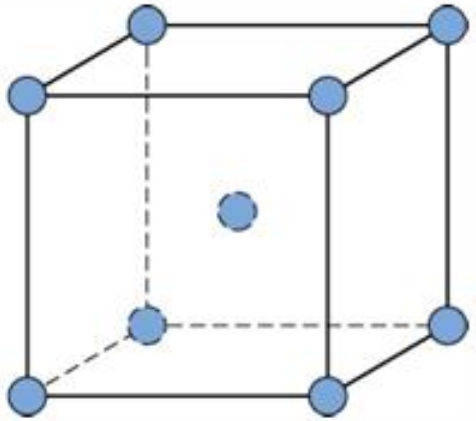
- Atoms touch each other along cube diagonals.
  --Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

ex: Cr, W, Fe (α), Tantalum, Molybdenum

- Coordination # = 8





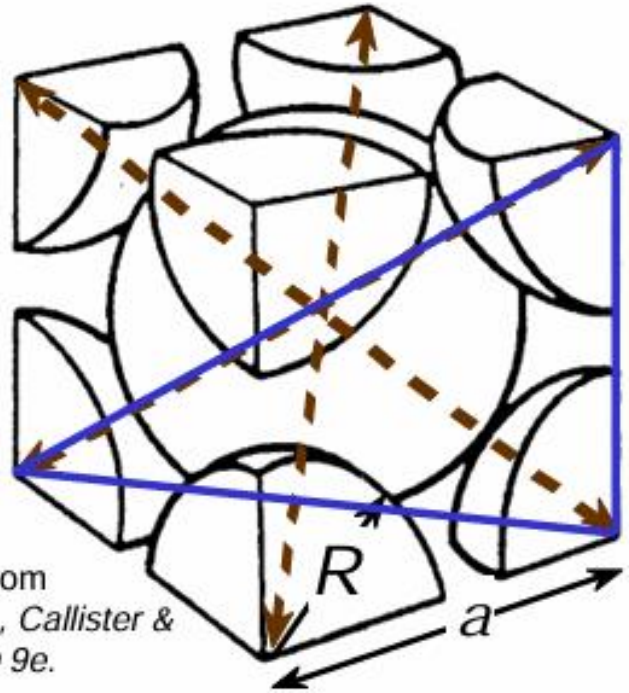


Adapted from Fig. 4.1,
Callister & Rethwisch 9e.

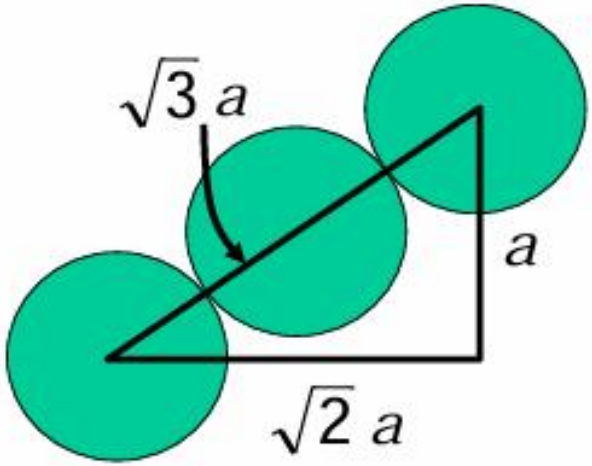
2 atoms/unit cell: 1 center + 8 corners x 1/8

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from Fig. 4.1(a), *Callister & Rethwisch 9e.*



Close-packed directions:
length = $4R = \sqrt{3}\,a$

$$\text{APF} = \frac{\left(\frac{\text{atoms}}{\text{unit cell}}\right)\left(\frac{\text{volume}}{\text{atom}}\right)}{\left(\frac{\text{volume}}{\text{unit cell}}\right)} = \frac{2\left(\frac{4}{3}\pi\left(\frac{\sqrt{3}a}{4}\right)^3\right)}{a^3}$$

$$APF = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell = Number of atoms X Volume of atom

for BCC Number of atom = close-packed directions =

Contains 1 atom+ 8 x 1/8 = 2 atom/unit cell

$$but\ 4r = a\sqrt{3}$$

Volume of atom = $\frac{4}{3}\pi(a\frac{\sqrt{3}}{4})^3 = \frac{4}{3}\pi\left(\frac{4r}{\sqrt{3}}\right)^3$

Volume of Unit cell = $a^3 = (\frac{4r}{\sqrt{3}})^3$

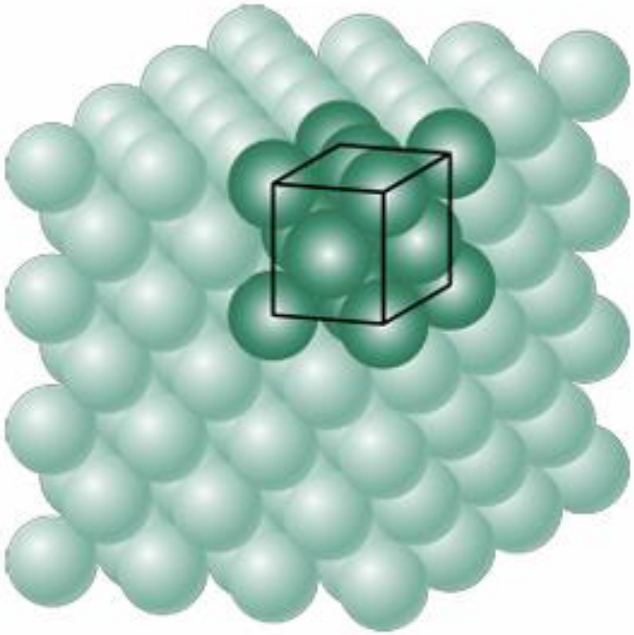
$$APF\ for\ BCC = \frac{2 \times \frac{4}{3}\pi\frac{a^3}{4^3}\sqrt{3}^3}{(a)^3} = 0.68$$

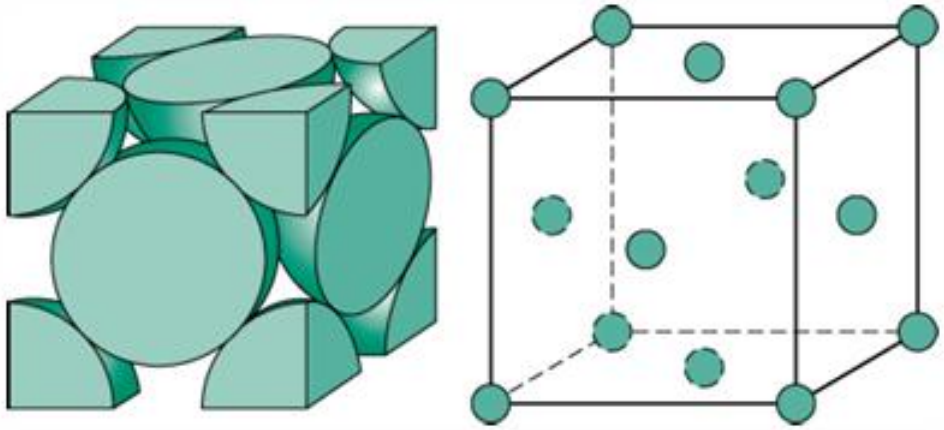
Face Centered Cubic Structure (FCC)

- Atoms touch each other along face diagonals.
  --Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

ex: Al, Cu, Au, Pb, Ni, Pt, Ag

- Coordination # = 12



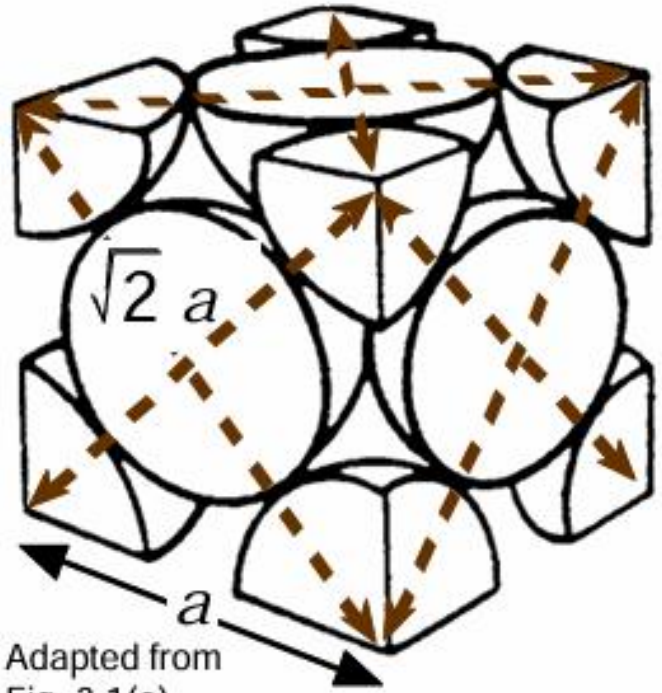


Adapted from Fig. 3.1, *Callister & Rethwisch 9e.*

4 atoms/unit cell: 6 face x 1/2 + 8 corners x 1/8

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74
  maximum achievable APF



Adapted from Fig. 3.1(a), *Callister & Rethwisch 9e.*

Close-packed directions:
length = $4R = \sqrt{2}\,a$

Unit cell contains:
6 x 1/2 + 8 x 1/8
= 4 atoms/unit cell

$$\text{APF} = \frac{\dfrac{\text{atoms}}{\text{unit cell}} \cdot \dfrac{\text{volume}}{\text{atom}}}{\dfrac{\text{volume}}{\text{unit cell}}} = \frac{4\left(\frac{4}{3}\pi\left(\frac{\sqrt{2}a}{4}\right)^3\right)}{a^3}$$

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

Volume of atoms in unit cell = Number of atoms X Volume of atom

for FCC Number of atom = close-packed directions =

Contains 6 x 1/2 atom+ 8 x 1/8 = 4 atom/unit cell

$$but\ 4r = a\sqrt{2}$$

Volume of atom = $\frac{4}{3}\pi(a\frac{\sqrt{2}}{4})^3 = \frac{4}{3}\pi\left(\frac{4r}{\sqrt{2}}\right)^3$

Volume of Unit cell = $a^3 = (\frac{4r}{\sqrt{2}})^3$

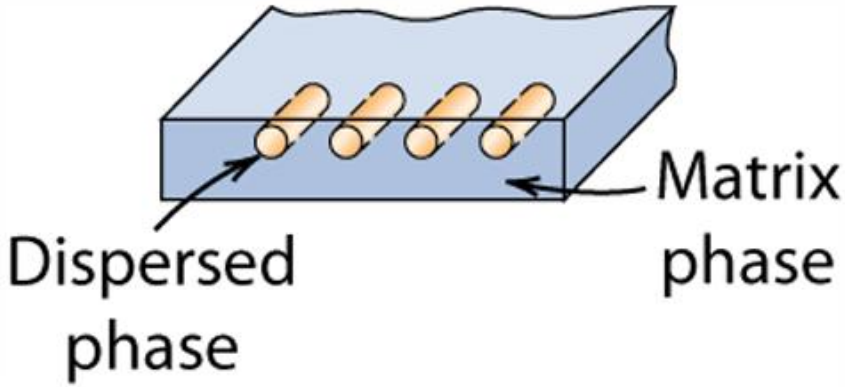
$$APF\ for\ FCC = \frac{4 \times \frac{4}{3}\pi\frac{a^3}{4^3}\sqrt{2}^3}{(a)^3} = 0.74$$

Terminology/Classification

- Phase types:

  -- Matrix Material Phase - is continuous

  -- Dispersed material Phase- is discontinuous and surrounded by matrix



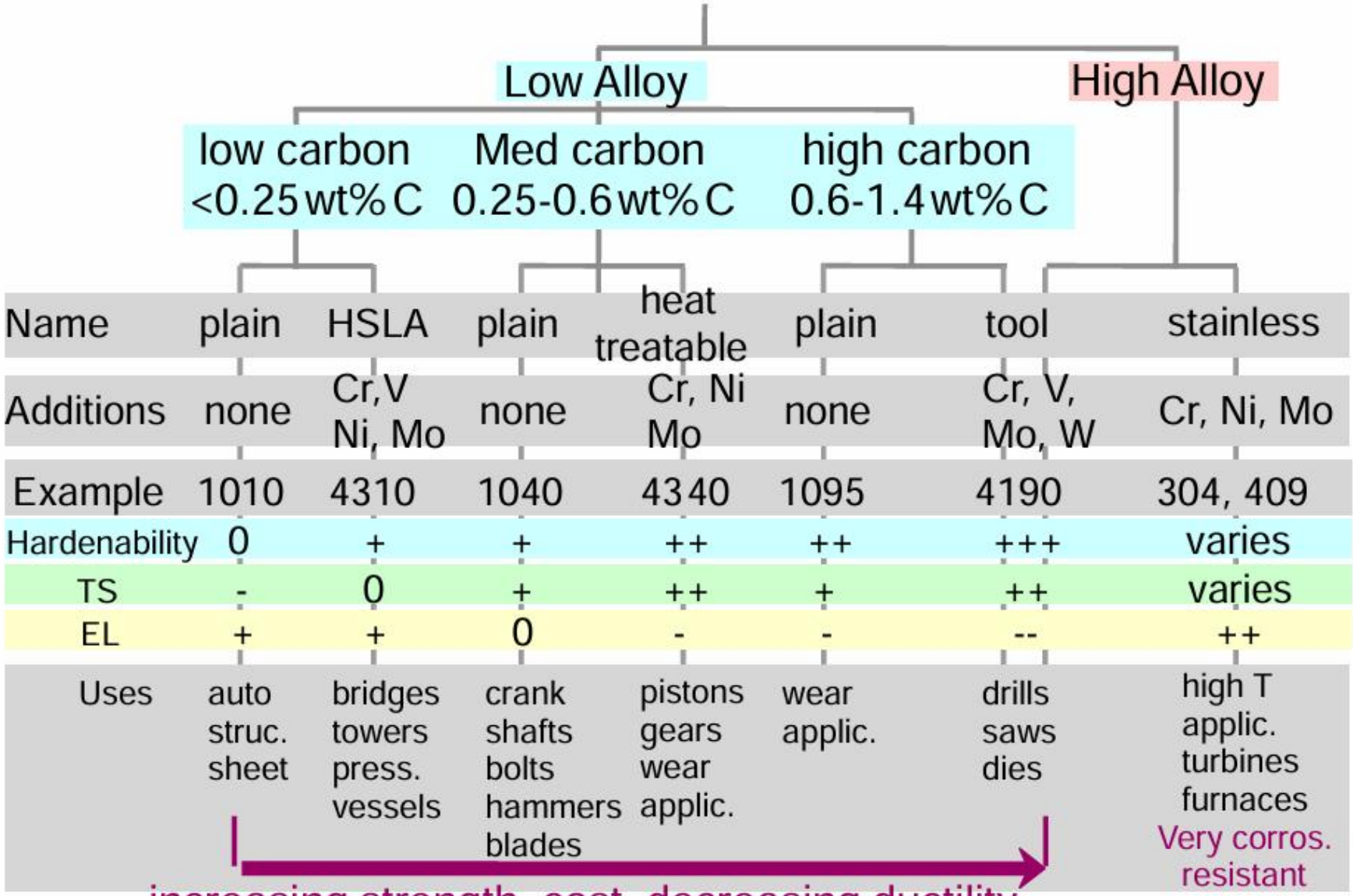
Adapted from Fig. 16.1(a), *Callister & Rethwisch 9e.*

The matrix is basically a homogeneous material in which a dispersed material of the composite is embedded. It is completely continuous.

The matrix provides a medium for binding and holding reinforcements together into a solid. It offers protection to the reinforcements from environmental damage, serves to transfer load, and provides finish, texture, color, durability and functionality.

The dispersed material or reinforcement is discontinuous and provides strength and stiffness

Steels



Based on data provided in Tables 13.1(b), 14.4(b), 13.3, and 13.4, *Callister & Rethwisch 9e.*

Thermal Processing of Metals

Annealing is a heat treatment process that changes the physical properties of a material to increase ductility and reduce the hardness to make it more ductile.

Stress Relief Annealing is a heat-treatment procedure of casts with the aim to reduce inner stress within the castings through heating and slow cooling-down.

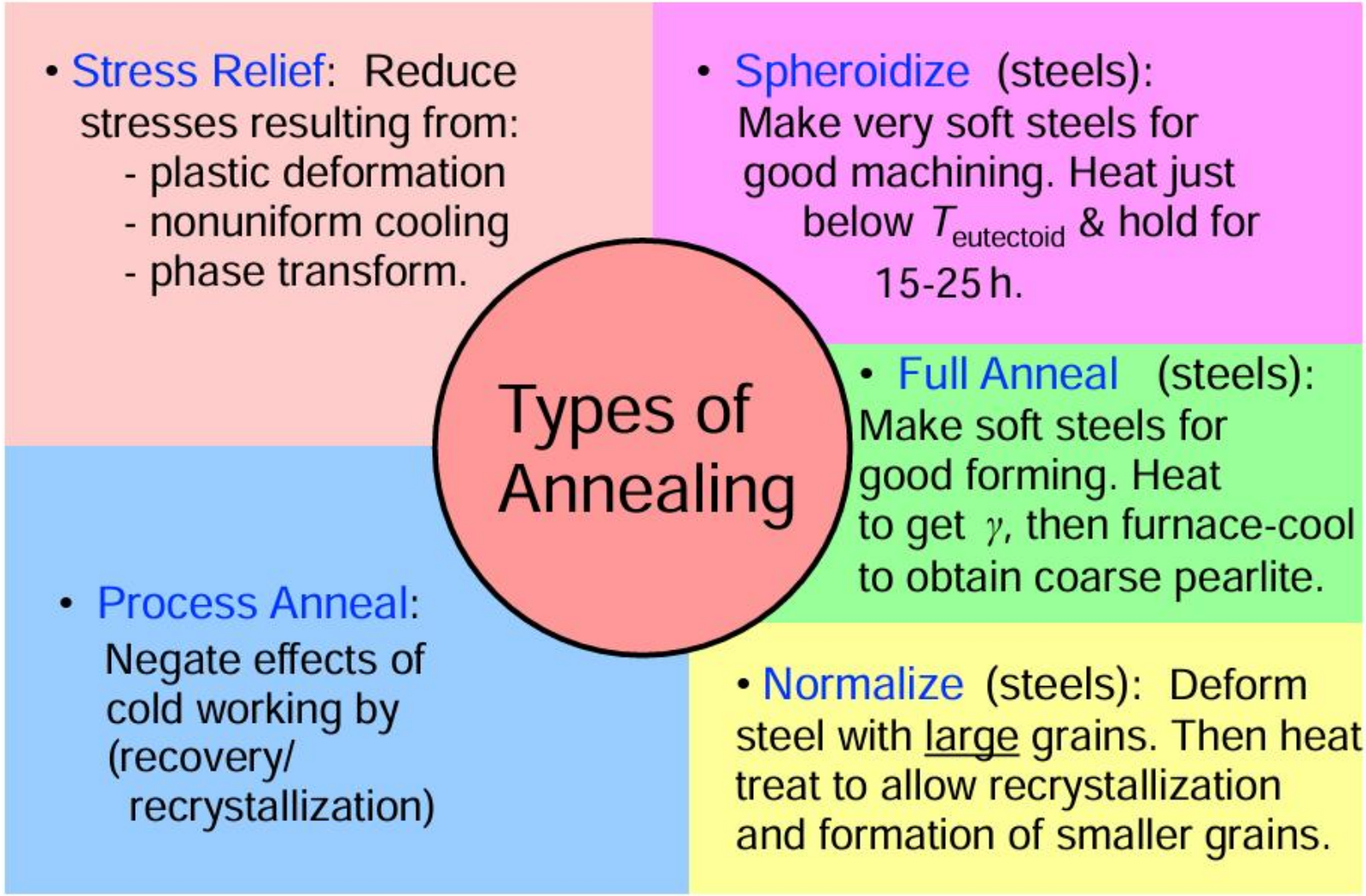
Spheroidize Annealing is a type of annealing process involving controlled heating and cooling to produce a microstructure consisting of spherical (spheroidal) carbides in a ferrite matrix.*

Process Annealing the process of softening steel by heating it to a temperature near but below the transformation (Solid to Liquid) range and then cooling slowly.

Normalizing is a heat treatment process in which steel is heated about its recrystallization temperature and slow cooled in air

Full annealing is the process by which distorted cold-worked lattice structure is changed back to one that is strain-free through the application of heat.

Thermal Processing of Metals



ECONOMIC CONSIDERATIONS

- A product must make economic sense:
  -- the price must be attractive to customers
  -- it must return a sustainable profit to the company
- To minimize product cost materials engineers must consider three factors
  -- component design
  -- material selection
  -- manufacturing techniques
- Other significant factors include labor & fringe benefits, insurance, and profit

COMPONENTS OF "GREEN DESIGN"

- **Reduce** – redesign the product to use less material
  example: PET bottles with thinner walls



Christopher Steer/iStockphoto

- **Reuse** – fabricate the product of a material that can reused
  example: refillable bottles and shipping containers
  example: grind up old tires for use as mulch

- **Recycle** – reprocess the material into a new product
  example: convert PET bottles to carpet fibers